

Kontena: A Developer-friendly Container and Microservices Platform

Kontena is driving innovation in container platform technology by accelerating projects and taking container-based release automation to the next level. Kontena developers can leverage multiple cloud providers into one secure virtual environment which means no lock-in, apart from many other uses.



Containerisation has been a hot topic in the industry for a long time now. Basically, containerisation is an OS level virtualisation technique for deploying and running distributed applications without launching separate VMs for a single application. Kubernetes is very popular for deploying, scaling and maintaining container based applications. But it requires a lot of manpower to install and maintain it. It is also complex to start with. Kontena is another open source system whose ultimate goal is to build a DevOps-friendly platform. It serves the same purpose that Kubernetes does, but in an easier and more manageable way so that a typical software developer can program and manage containers and micro systems.

Kontena

Kontena is an open source project available in the GitHub Repository that helps us to run and manage containerised applications on a cluster. Figures 1 and 2 demonstrate its architecture. The Kontena system has a number of nodes and a master node, which controls the Kontena nodes. Kontena nodes are servers or VMs, which run container apps and the

Kontena master is the controller for all the nodes.

To build an application, we require Kontena Service Definition, which includes details about the container image, scaling, networking details and other attributes. These services can be linked together to create the desired architecture. Here, each service will automatically generate an internal IP address and DNS, so that they can communicate internally.

Architecture

One of the best things about Kontena is its simple architecture, which makes it easier to grasp and so you can quickly start using it. With the Kontena CLI, we can do all the administrative stuff. Currently, there is no GUI for Kontena, which makes the code simpler and manageable.

The architecture consists of four layers:

- The authentication layer (external).
- Masters, which will manage each grid.
- Grids, which are logical abstractions. Nodes will recede into a grid, which provides actual compute power.
- Services that will run on the grid.